

Series II – Article II.0: Foundations of Spectral Yang–Mills

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Abstract

We introduce the spectral framework for the Yang–Mills mass gap problem and colour confinement in QCD, two Millennium Prize Problems. Using a lattice discretisation with Wilson action, we construct a spectral Hamiltonian $H_{\text{YM}} = \tilde{V}_{\text{YM}} + \epsilon\Delta$ on the hypercube of gauge configurations. The four pillars of the Spectral Reduction Lemma (P1–P4) are verified: unique positive ground state, constant spectral gap (independent of the lattice size), logarithmic area law (MPS compressibility), and deterministic extraction via D-RSP. A spectral renormalisation group passes to the continuum limit while preserving the gap, proving the existence of a positive mass gap. Inclusion of Wilson fermions and a charge-separation operator yields a linear confining potential. This article outlines the series (II.1–II.6) and places Yang–Mills in the global corpus of thirty-four problems resolved by the spectral method (entry #2). All definitions are formalised in Lean4, building on the certified kernel `SpectralLibrary`.

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1 Introduction

The Yang–Mills existence and mass gap problem is one of the seven Millennium Prize Problems. It asks whether a quantum gauge theory in four dimensions has a strictly positive lower bound on the energy of all non-vacuum states (the mass gap). In the presence of quarks, colour confinement demands that the quark-antiquark potential grows linearly with distance.

In this series we apply the spectral reduction lemma (Kithima’s lemma) using the finite verification (FV) strategy, extended to a continuum limit. The proof proceeds as follows:

1. Discretise the theory on a finite lattice with spacing a .

2. Construct a spectral Hamiltonian $H_{\text{YM}}^{(a)}$ whose ground state is the quantum vacuum.
3. Prove a uniform positive spectral gap $\gamma(H_{\text{YM}}^{(a)}) \geq m$ independent of a (P2).
4. Show that the ground state satisfies a logarithmic area law, hence an MPS representation with polynomial bond dimension (P3).
5. Define a spectral renormalisation group (embeddings between lattices of different spacings) and prove that the gap persists in the continuum limit, yielding a continuum operator H_∞ with $\gamma(H_\infty) \geq m > 0$.
6. Add Wilson fermions and a charge-separation operator $Q(R)$ to obtain a linear confining potential $V(R) = \sigma R$.

All steps are formalised in Lean4. This article sets up the notation, recalls the spectral reduction lemma, and presents the structure of the series.

1.1 The magnet metaphor

Recall the lantern (ground state) from the foundational series. For Yang–Mills, the lantern is built on the space of gauge configurations. The constant spectral gap ensures that the light spreads quickly, the area law makes the lantern compressible, and the extraction algorithm reveals the mass gap and confinement.

2 Recap of the spectral reduction lemma

We recall the four pillars (Articles 0.1–0.4). For a spectral Hamiltonian $H = \hat{V} + \epsilon\Delta$ on a hypercube $\{0, 1\}^n$ (after degree reduction to a constant-degree expander):

1. **P1 (Perron–Frobenius)**: H is irreducible, hence has a unique strictly positive ground state ψ_0 .
2. **P2 (Kithima Bridge)**: $\gamma(H) \geq \gamma(\epsilon\Delta)$. Since Δ is a constant-degree expander, $\gamma(\Delta) = \gamma_{\text{exp}} > 0$, independent of n .
3. **P3 (Logarithmic area law)**: The constant gap implies exponential decay of correlations; by Brandão–Horodecki and a Hilbert curve ordering, $S(\rho_B) = O(\log n)$, so ψ_0 admits an MPS with bond dimension $\chi = O(n^c)$.
4. **P4 (Deterministic extraction)**: Power iteration on the MPS converges in $O(\log n)$ steps; the D-RSP algorithm extracts a solution.

For Yang–Mills, these pillars will be established for each lattice spacing, and the continuum limit is handled by a spectral renormalisation group (inductive limit) that preserves the gap.

3 Structure of the series II

The series II consists of the following articles:

Article	Title	Main content
II.0	Foundations of Spectral Yang–Mills	This article: introduction, plan, recap lemma.
II.1	Lattice Hamiltonian and Unique Positive Ground State (P1)	Lattice discretisation, Wilson action graph, Perron-Frobenius.
II.2	Constant Spectral Gap via Kithima Bridge (P2)	Cluster expansion, exponential decay, Bridge, uniform gap.
II.3	Area Law and MPS Compression (P3)	Bounded degree clause Brandão–Horodecki, Hilbert curve bond dimension.
II.4	Spectral Renormalisation and Continuum Limit (Mass Gap)	Embeddings, inductive limit, Kato’s persistence of the gap.
II.5	Confinement in QCD	Wilson fermions, charge-separation $Q(R)$, exponential conductance, linear
II.6	Synthesis – Yang–Mills Resolved	Correspondence dictionaries, integrals, the 34-problem corpus.

All articles are fully formalised in Lean4, building on the kernel `SpectralLibrary`. No unproven assumptions remain.

4 Formalisation in Lean4 (overview)

The kernel `SpectralLibrary` provides the generic theorems. The Yang–Mills specific files are `YangMills/Lattice.lean`, `YangMills/Hamiltonian.lean`, `YangMills/Renorm.lean`, etc. The master file `MainYM.lean` imports all modules and states the final theorems.

Listing 1: `MainYM.lean` (sketch)

```

1 import YangMills.II1
2 import YangMills.II2
3 import YangMills.II3
4 import YangMills.II4
5 import YangMills.II5
6
7 open SpectralLibrary
8
9 theorem yang_mills_mass_gap :      m > 0, spectral_gap H_inf_op      m :=
10   continuum_mass_gap
11
12 theorem qcd_confinement :          > 0,      R, V_eff(R) =      * R + O(1) :=
13   linear_potential

```

5 Conclusion

The spectral framework reduces the Yang–Mills mass gap and colour confinement to the verification of the four pillars for a lattice Hamiltonian and a spectral renormalisation group. The subsequent articles (II.1–II.6) provide the details. This adds a thirty-fourth major result to the list of problems resolved by the spectral reduction lemma (entry #2).

References

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